# Why the added road did not improve travel time

Decision: do not present this design as a travel-time improvement. In three matched-seed comparisons, the alternative was effectively unchanged relative to closure and did not recover the closure loss.

**+7.530 s**
Closure minus baseline

**+0.098 s**
Alternative minus closure

**6 / 934**
Observed new-road trips / run

## 01 A controlled comparison, with explicit denominators

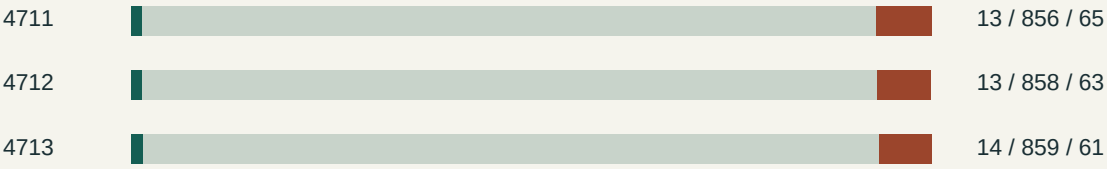
One fixed synthetic population of 1,037 people; three scenarios per seed (4711, 4712, 4713); iterations 0-9. Each run has 3,623 trips, including 934 car trips. Pairing uses person + trip number, anchored to the before-scenario departure in [0, 30 h). All 934 car keys pair; no mode switches in these comparisons. [Q01-Q03]

| Seed | Baseline mean (s) | Closure mean (s) | Alternative mean (s) | Alt. - closure (s) |
|---|---|---|---|---|
| 4711 | 235.703 | 243.409 | 243.526 | +0.117 |
| 4712 | 235.697 | 243.221 | 243.318 | +0.097 |
| 4713 | 235.878 | 243.239 | 243.318 | +0.079 |

The three-seed average is descriptive. The same population is reused; 2,802 paired records are not 2,802 independent travellers. A +0.098 s difference is not evidence of meaningful deterioration or statistical significance.

## 02 Small gains are offset by losses

Alternative versus closure: median change is 0 s and p95 is +1 s for every seed. Positive means slower. Bins use faster <= -1 s, unchanged |delta| < 1 s, slower >= +1 s. [Q03-Q04]




Bars: faster / unchanged / slower. Seed 4711 saves 205 trip-seconds but adds 314, leaving +109 / 934 = +0.117 s. Seeds 4712 and 4713 leave +91 and +74 trip-seconds. One trip is +109 s in each seed; its route-level cause is not established here.

Compared with the original baseline, the alternative remains +7.628 s slower on average across seeds. Low median change therefore does not mean the closure impact was resolved. [Q03-Q04]

# The explanation, and its limits

## 03 Evidence supporting low route attractiveness and limited reach

| Evidence | Interpretation |
|---|---|
| Same-endpoint free-flow costs [Q09]<br>Existing shortest: 20.88 s<br>Full new corridor: 42.74 s / 356.16 m | The new corridor is about 2.05 times the existing minimum. The network minimum is unchanged in both directions after adding it. This supports a lack of endpoint-to-endpoint time advantage; it is not an observed trip-time prediction. |
| Observed usage [Q07]<br>6 / 934 car trips (0.64%)<br>13 of 32 new directed links used | The new road reaches few recorded trips. Each seed has 17 complete new-link traversals across six trips, not 17 users. Partial entry/exit links are excluded; absence of a complete traversal is not proof of no use. |
| The affected baseline cohort [Q06]<br>223-226 trips used the closed links | Their mean alternative-minus-baseline change remains +30.37 to +30.89 s. Adding the road changes this cohort by only +0.15 to +0.23 s versus closure; it does not restore the original connection. |

Design context: the 356 m corridor was chosen for geometric feasibility before traffic outcomes, with 30 km/h, one lane per direction and capacity 900 vehicles/hour. A footprint-feasible path is not a travel-time-optimised path. Closed links and the new corridor have different endpoints; the direct cost test above deliberately uses the same endpoints. [case_spec.json, Q09]

## 04 What the remaining diagnostics can and cannot tell us

**Observed:** the six new-road users are +5.50 to +5.67 s slower on average versus closure. This is a subgroup selected after the intervention, so it is not a causal estimate of the road's effect on users. Other trips also change. [Q06]

**No broad delay relief is visible on comparable links:** the largest absolute mean-time change between alternative and closure is about 0.014 s (links 1871 and 9344). At seed 4711 both move from 72 to 73 complete traversals and from 10.000 to 10.014 s / 7.000 to 7.014 s. Changed geometry and unobserved links are excluded; changing traffic composition still matters. [Q08]

**Time windows:** alternative minus closure is +0.068 s in 07-09, -0.022 s in 16-18, and +0.107 to +0.166 s otherwise. These descriptive windows do not demonstrate peak congestion relief. [Q05]

## 05 Next experiments, in order

**1. Isolate topology from the road itself.** Run closure + identical link splits with the new corridor disabled; compare it with the full alternative using matched seeds. This separates junction/splitting effects from added connectivity.
**2. Test route attractiveness.** Inspect actual paths of the six users and the +109 s trip; then test a predeclared speed/capacity grid and demand-linked endpoint choices. Keep plans and seeds fixed.
**3. Test stability and realism.** Increase iterations and check score/route stability, expand seeds, then calibrate against observed demand/counts before making real-world claims. No convergence or policy-benefit claim is made now.

**Audit trail.** Q01-Q10 map to sql/queries/01_...sql through 10_...sql; aggregate CSVs and source SHA-256 values are in portfolio/evidence/sql/. SQL retains missing/invalid values as NULL and real zero as zero. Nine-run metrics and pair distributions match the existing Python analysis; event parsing is shared, not independently reimplemented. See docs/ANALYTICS.md for definitions and docs/CONTRIBUTIONS.md for upstream and AI-assisted boundaries.